AI

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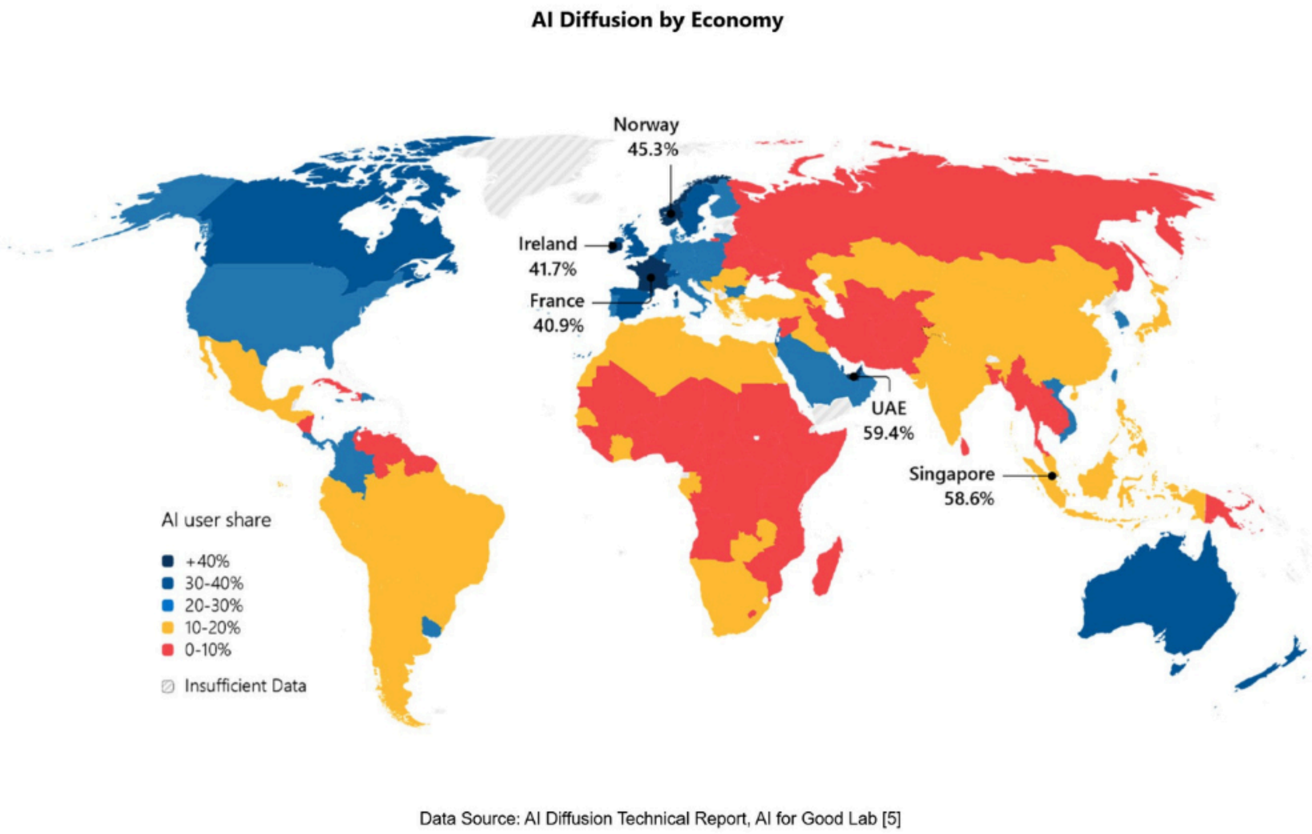
AI Diffusion Report: Mapping global AI adoption and innovation

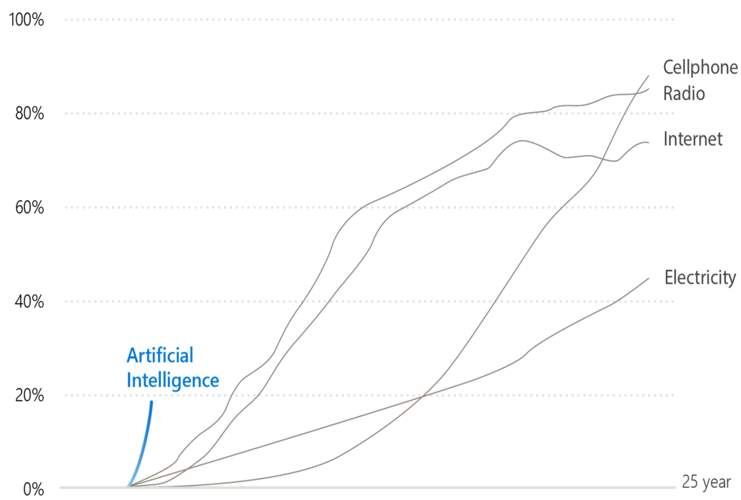
By Jana Cvetko



Low adoption, high potential in Ukraine: At **9.1%**, Ukraine reflects the impact of war-related disruptions but holds significant potential for rapid growth with targeted investments.

The AI Economy Institute, Microsoft's flagship think tank released its [AI Diffusion Report], offering comprehensive view of where artificial intelligence is being used, developed, and built globally. The report provides country-level estimates of AI adoption, insights into innovation hubs, AI skills trends, and the critical role of digital infrastructure in enabling equitable access to AI.

Artificial intelligence is the next great general-purpose technology—and the fastest-spreading technology in human history. In less than three years, **more than 1.2 billion people have used AI tools,** a rate of adoption faster than the internet, the personal computer, or even the smartphone. Yet, similar to other general-purpose technologies before, its benefits are not spreading evenly. AI use in the **Global North** is roughly double that in the Global South. Without focused effort, this gap will define who benefits from AI for decades to come.

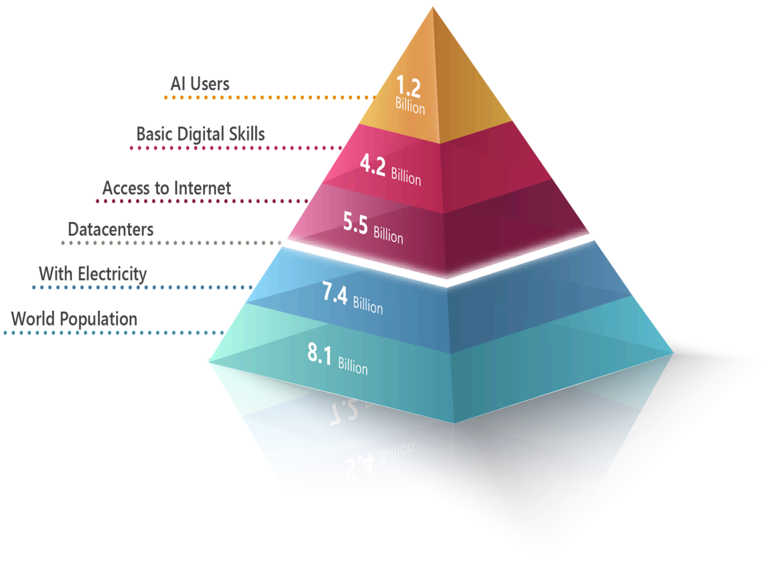


Comparison of AI diffusion to other earlier technologies. Source: Microsoft AI Diffusion Report 2025.

Three forces driving AI transformation

AI's rapid diffusion is powered by three interconnected forces: **frontier builders**—researchers and model creators pushing the limits of intelligence; **infrastructure builders**—engineers and institutions scaling breakthroughs through compute, connectivity, and skills; and **users**—individuals and organizations applying AI to learn, create, and solve real-world problem.

The building blocks of AI



World population compared to the share of the population with access to electricity, individuals using the Internet in 2024, and Internet users with basic information and data literacy skills. Source: AI Diffusion Report 2025.

AI adoption rises where strong economic foundations exist and lags where they do not. At the heart of this divide are the essential building blocks: **electricity** powering devices and data centers; **data centers** forming the backbone of global networks; the **internet** connecting users and enabling data flow; **digital and AI skills** unlocking the ability to use and innovate; and **language**, which determines who can access AI and how it evolves.

"For Ukraine, technology is a lifeline. AI is already strengthening cybersecurity and helping businesses and organizations stay resilient. While overall AI adoption is low, strategic investment in connectivity, digital and AI skills, and language-inclusive solutions can accelerate growth and support recovery. By leveraging advanced technologies and lessons from more mature markets, Ukraine can rebuild faster, creating stronger, more innovative, and resilient communities in the future." **Leonid Polupan, Country Manager, Microsoft Ukraine and Baltics.**

Leonid Polupan, Country Manager, Microsoft Ukraine and Baltics

Key findings

- **Fastest adoption in history:** As a general-purpose technology, AI stands on the shoulders of three others—electricity, connectivity, and computing. Its adoption is fastest where these foundations exist, and slowest where they do not. Nearly four billion people—half the world—still lack the basics needed to use AI.
- **Uneven benefits:** AI adoption in the Global North is roughly twice that of the Global South, with the gap widening sharply in countries where GDP per capita falls below $20,000. In some Global North countries, more than half of the working-age population uses AI, while in parts of Sub-Saharan Africa and Asia, some of the least developed nations have adoption rates below 10%.
- **Language matters**: Nations where low-resource languages dominate—like Malawi or Laos—show lower adoption even after adjusting for GDP and internet access.
- **AI Diffusion leaders:** the UAE (59,4%), Singapore (58,6%), Norway (45,3%) and Ireland (41,7%) —stand

out as *leaders in AI Adoption*, proving that strong access to technology, education, and policy coordination can drive rapid adoption even without frontier-level model development or data centers.

- **Low adoption, high potential in Ukraine:** At **9.1%**, Ukraine reflects the impact of war-related disruptions but holds significant potential for rapid growth with targeted investments.
- **Frontier narrowing:** From a **frontier builder** perspective, the number of AI models continues to rise, while the performance gap between them keeps narrowing. The U.S., led by OpenAI's GPT-5, remains at the frontier, with China trailing by less than six months. Only seven countries—the U.S., **China,** France, South Korea, the U.K., Canada, and Israel—rank among the top 200 models, and the distance between the frontier (U.S.) and the last of these (Israel) is now just 11 months.
- **Infrastructure concentration:** From an **infrastructure builder** perspective, the U.S. and China together host 86% of global data center capacity, underscoring how concentrated the foundation of AI remains.

"*We continue to stand with Ukraine—not only through initiatives like AI Skills Navigator, LinkedIn Learning, and Microsoft for Startups, but also by democratizing access to AI with tools such as free Copilot Chat and ensuring a secure, safe and reliable AI use through our cybersecurity solutions. By combining deep technology expertise, regional and global experience, and our strong partner ecosystem, we can accelerate AI adoption and deliver real impact – helping build the foundation for a resilient, innovative future,"* **Polupan concluded.**

Artificial intelligence is the next great general-purpose technology—and the fastest-spreading in history. Yet, like electricity and connectivity before it, its benefits are not evenly distributed. Closing the adoption gap will require coordinated efforts across infrastructure, skills, and language accessibility.

About the AI Economy Institute

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Top image: The Microsoft AI Diffusion Report 2025 highlights global AI adoption trends, regional gaps, key drivers, and calls for investment in infrastructure, skills, and secure, inclusive AI use. Image courtesy of Microsoft.

AI for education